



Technical Service Bulletin

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Variable Geometry Turbocharger Actuator Installed Incorrectly

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- B6.7 CM2350 B121B
- L9 CM2350 L116B
- X15 CM2350 X114B
- X15 CM2350 X116B

Issue Summary

Symptom:

- Fault Code 3382
- Fault Code 3361
- Fault Code 3389
- Air Handling Performance Test did **not** pass

Root Cause:

- Variable geometry turbocharger actuator installed incorrectly

Verification

Verify one of the fault codes listed are active or inactive with more than one count and the Air Handling Performance Test did **not** pass for one of the messages in Table 1.

Table 1, Air Handling Performance Test and Log Message Reference	
Insite Message	Repair Decision message in test log
The VGT is causing exhaust pressure to dither.	VGT Stuck
An exhaust manifold leak has been detected.	Exhaust Leak

Table 1, Air Handling Performance Test and Log Message Reference

Insite Message	Repair Decision message in test log
A position error has been detected with the VGT.	VGT Global Position Error

- Find the did **not** pass log stored on the computer that ran the test, located at C:\Intelect\APT.
- Open the log, as shown in Figure1, and confirm that the ESN (1) is correct and that the Repair Decision (2) matches Table 1.
- If the VGT Minimum Position Parameter (3) shown in Figure 1 is greater than 4 percent, continue to the Resolution section.

Air Handling Performance Test		
Guidanz Job:NA		
Date:05/20/2022		
Time:09:20:58		
Engine Serial Number:80206333		1
ECM Code:HD10362.09		
Engine Hours:007019:51:45		
Engine Distance:282210.1		
EGR Delta Pressure Autozero Offset:0.0		
EGR Valve Position Offset:-107.0		
Repair Decision:VGT Stuck		2
Abort Reason:No Abort		
Fault Code	Counts	Status
	3382	10 Active
APT Rationality Check Log	Value	Unit
VGT Minimum Position	7.2	percent 3
VGT Maximum Position	100	percent
EGR Minimum Position	0	percent
EGR Maximum Position	100.7	percent
Intake Air Throttle Minimum Pos	0	percent
Intake Air Throttle Maximum Pos	99.8	percent
10r00488		

Figure 1, Air Handling Performance Test Log Screenshot.

Resolution

See corresponding Service Manual. Reference Procedure 010-134 in Section 10 and perform the following:

- Remove the actuator and check sector gear travel.
- If sector gear travel is within specification, Perform Actuator Install and Calibrate Steps.
- If Install and Calibrate completes successfully, run the Air Handling Performance test to validate the repair.

Document History

Date	Details
2024-8-28	Module Created

Last Modified: 09-Sep-2024
